GC500

Auto and Manual Start Genset Controller

HIGHLIGHTS »

- **1-Ph/3-Ph genset voltage and frequency sensing**
- **3 analog inputs, 5 digital inputs and 6 digital outputs**
- **Deep sleep mode for prolonged battery life**
- **Remote start/stop feature to control the start/stop action of genset remotely**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**

GC500 Controller

SEDEMAC GC500 is a manual and auto-start genset controller with user-friendly HMI and full graphics LCD.

It offers engine safety monitoring, genset voltage and frequency measurement (true RMS).

Fully configurable analog, digital inputs and outputs are provided for various features.

Remote Start/Stop feature is provided, through which one can control the start/stop actions of the genset remotely.

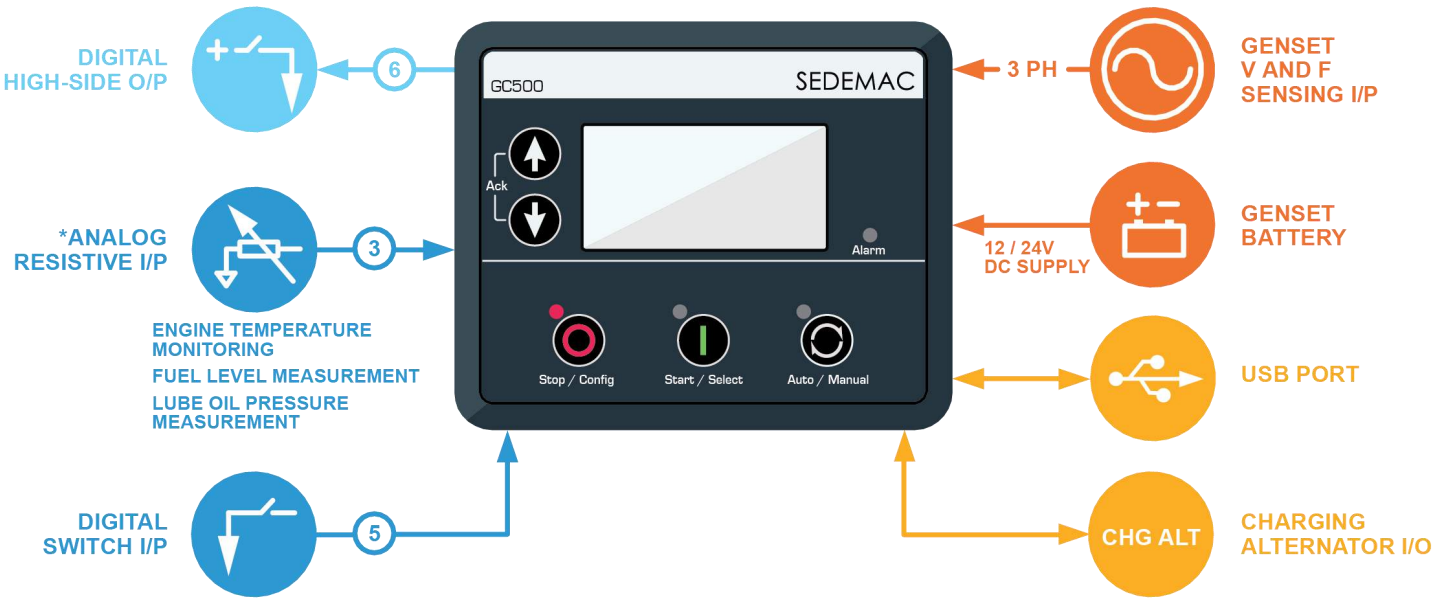
Deep Sleep Mode is a useful feature being provided to prolong the battery life by suspending the normal functions of the controller when DG is off.

The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

An event log upto **30 events** is provided for data analysis.

GC500

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics back-lit LCD display with power save mode
- Monitors voltage and frequency of generator (1-Ph/3-Ph)
- Monitors engine safety parameters through analog resistive sensors and digital switch inputs (e.g. lube oil pressure, engine temperature, fuel level)
- Commands start relay, fuel, stop relay, alarm hooter and more
- Supports the battery charging alternator I/O interface
- Commands the genset to start or stop through the preconfigured remote start/stop input
- Versatile software capable of handling "events" during operation; Event log for 30 events for analysing the data
- Incremental counters for engine starts, stops & running hours
- Communicates with PC via Windows based configuration utility called **SEDEMAC-Smart Config.** through a USB port
- Supports end-of-line configuration at customer's site
- Offers multiple level passwords to avoid tampering in the configuration
- Access to each parameter through SEDEMAC's PC software & control keys of GC500 controller

Ordering Information:

GC500 Controller SM0003919

SPECIFICATIONS »

Environmental testing standards

Operating/Storage temperature
-20˚C to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X,Y and Z axes for 8-500 Hz, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V - 32V DC continuous; power transients as per ISO 7637 - 2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @ 12V or 24V DC

Crank-Start Dropout Survival Period
50 mS for a drop from 12V DC

Charging Alternator Interface
0.25A @ 12V DC; 0.125A @ 24V DC
Diagnostic voltage measurement

Digital Output
6x 0.75A high-side drivers configurable for pre-heat, hooter, etc.

Digital Input
5 switch-to-ground inputs configurable for lube oil, temp, fuel level, etc.

Analog Input
3 resistive inputs (10Ω - 1000Ω); Fully configurable for various sources

Genset Voltage Measurement
20V-350V AC RMS @ 5 - 75 Hz for Ph to N

Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

GC800

E-governing Genset Controller

HIGHLIGHTS »

- **Auto, manual and remote start/stop modes of operation**
- **Built-in electronic governor module and offers multi-profiles (10 profiles) in a single controller**
- **MODBUS over RS485 and CAN port for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Backlit and full graphics display with power saving feature**

GC800 Controller

SEDEMAC's GC800 controllers are modern AMF genset controllers with user- friendly interface and full LCD graphics.

These controllers offer voltage and frequency measurement for mains as well as for genset and electrical load measurement (true RMS). Prior to that they also offer engine safety monitoring feature

Deep Sleep Mode is a useful feature being provided to prolong the battery life by suspending the normal functions of the controller when DG is off.

MODBUS over **RS485** and **CAN ports are** provided for supporting the remote communication.

The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

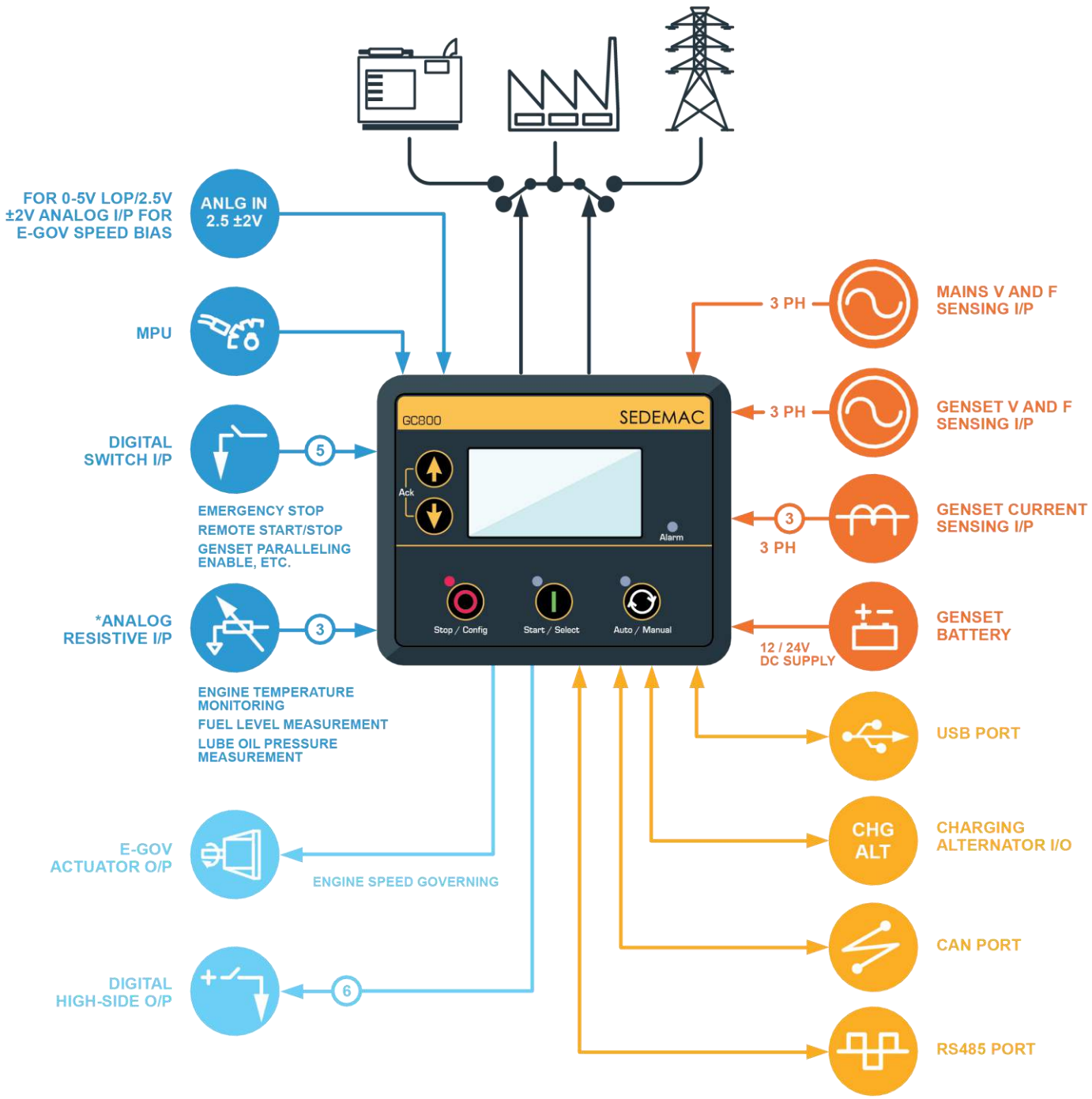
Real Time Clock (RTC) is provided for accurate event logging (100 counts) and data analysis.

Provision is there to Interface **Magnetic Pickup Unit** (MPU) to read the Engine RPM

In addition to above functionalities, GC800 controller also offers **Electronic Governing** function with the help of a rotary actuator. It supports **genset paralleling** through an analog input signal of 2.5V ±2V , which if not utilised can also be configured as a 0 - 5V analog sensor input.

GC800

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics back-lit LCD display with power save mode
- Supports Auto (AMF, remote start/stop and cyclic modes) and manual modes
- Monitors 1-Ph/3-Ph voltage, frequency and load current for generator and mains; monitors 1-Ph/3-Ph power factor for generator
- Offers ''Auto exercise mode'' (2 events) to start and stop the genset for preconfigured time
- Monitors engine safety parameters through analog resistive sensors and digital input switches (e.g. - lube oil pressure, engine temperature, fuel level)
- Controls start relay, fuel relay, stop relay, alarm hooter and more
- Supports the battery charging alternator I/O interface
- MPU interface for engine speed measurement
- Versatile software capable of saving "events" during operation; event log for 100 events with real time clock (RTC) stamps and engine run hour information
- GC800 has built-in electronic governor module which directly connects to SEDEMAC's rotary actuators* to guarantee speed governing performance as per ISO8528:G3 class
- Accepts target speed bias input (2.5V ± 2V) from an external load sharing module (LSM) for paralleling of gensets
- Totalizers for engine starts, running hours, kWh, kVAh, etc.
- MODBUS over RS485 and CAN ports for communicating with other devices
- Supports parameter configuration at customer's site
- Offers multiple level passwords to avoid tampering in the configuration
- Access to each parameter through PC software or control keys of GC800 controllers

Ordering Information:

GC800 Controller SM0007264

GC800 requires rotary actuator for E-governing

**Kindly refer SEDEMAC's datasheet of "rotary actuators for electronic speed governing" for interfacing an actuator with controller in governing application.*

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20˚C to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V - 32V DC continuous; power transients as per ISO 7637-2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @ 12V or 24V DC

Crank-start Dropout Survival Period
50 mS for a drop from 12V DC

Charging Alternator Interface
0.25A @ 12V DC; 0.125A @ 24V DC
Diagnostic voltage measurement

Digital Output
6 x 0.75A high-side drivers configurable for pre-heat, hooter, etc

Digital Input
5 switch-to-ground inputs configurable for lube oil, temp, fuel level, etc

Analog Input
3 resistive inputs (10Ω - 1000Ω); fully configurable for various sources
1 analog input (2.5V ± 2V) for E-Governor speed bias and paralleling or can be used as 0 - 5V analog sensor

E-governor Module Interface
Output for rotary actuator
(2.5V ± 2V) input for target speed bias; 2.5V for zero bias

Mains/Genset Voltage Measurement
20V – 350V AC RMS @ 5 – 75 Hz for Ph to N

Load Current Measurement
0A - 5A AC RMS for current transformer (CT) secondary

Magnetic Pickup Measurement
0.2V – 45V @ 10 Hz – 10 kHz

Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

GC1100

AMF Genset Controller

HIGHLIGHTS »

- **Auto, manual and remote start/stop modes of operation**
- **Offers multi-profile (10) in a single controller**
- **MODBUS over RS485 and CAN port for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Backlit and full graphics display with power saving feature**

GC1100 Controller

SEDEMAC's GC1100 controller is modern AMF genset controllers with user- friendly interface and full LCD graphics.

These controllers offer voltage and frequency measurement for mains as well as genset and electrical load measurement (true RMS). Prior to that they also offer engine safety monitoring feature.

Deep Sleep Mode is a useful feature being provided to prolong the battery life by suspending the normal functions of the controller when DG is off.

MODBUS over **RS485** and **CAN ports are** provided for supporting the remote communication.

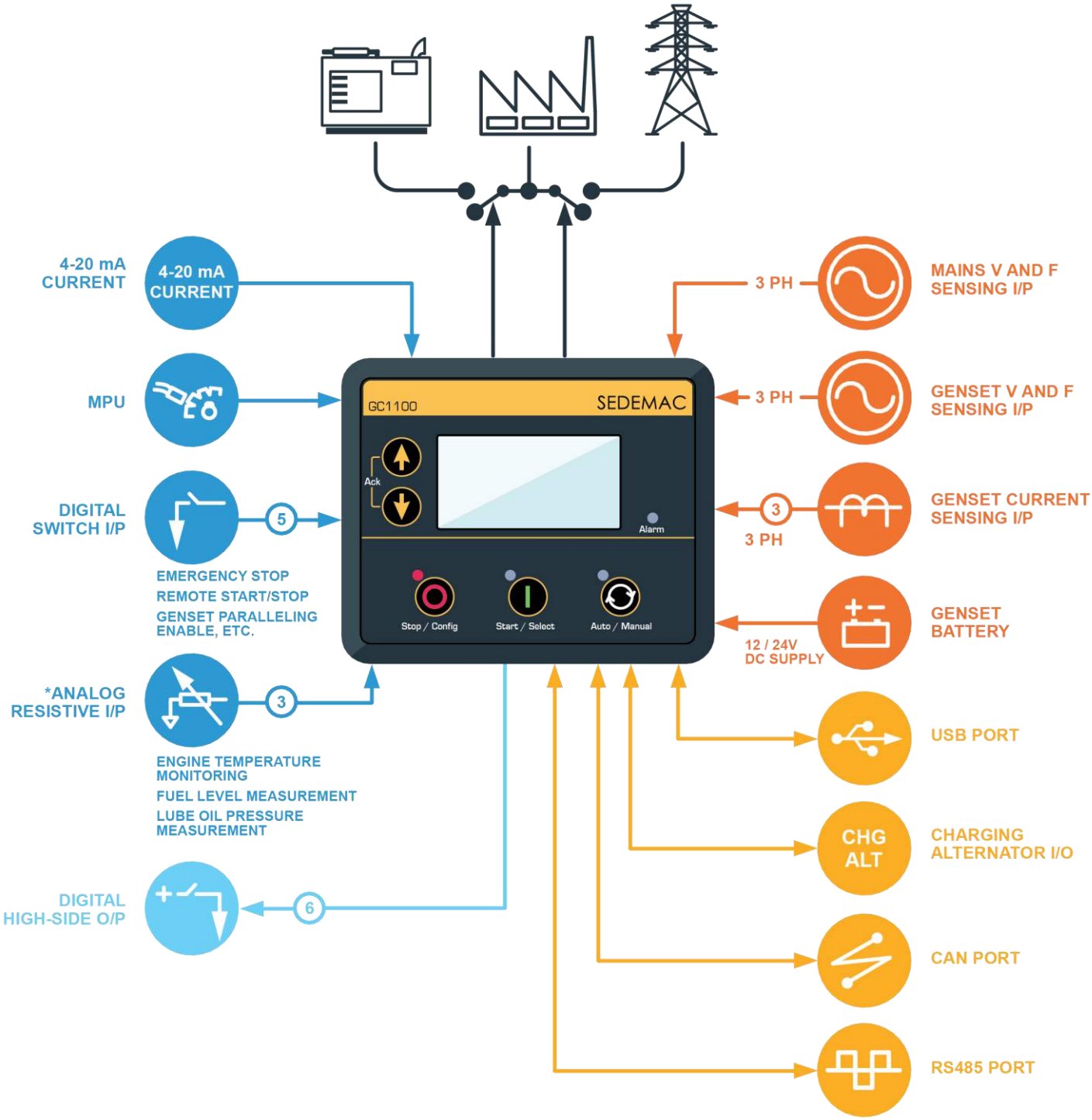
The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

Real Time Clock (RTC) is provided for accurate event logging (100 counts) and data analysis.

Provision is there to Interface **Magnetic Pickup Unit** (MPU) to read the Engine RPM.

GC1100

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics back-lit LCD display with power save mode
- Supports Auto (AMF, remote start/stop and cyclic modes) and manual modes
- Monitors 1-Ph/3-Ph voltage, frequency and load current for generator and mains; monitors 1-Ph/3-Ph power factor for generator
- Offers ''Auto exercise mode'' (2 events) to start and stop the genset for preconfigured time
- Monitors engine safety parameters through analog resistive sensors and digital input switches (e.g - lube oil pressure, engine temperature, fuel level)
- Controls start relay, fuel relay, stop relay, alarm hooter and more
- Supports the battery charging alternator I/O interface
- MPU interface for engine speed measurement
- Versatile software capable of saving "events" during operation; event log for 100 events with real time clock (RTC) stamps and engine run hour information
- Accepts target speed bias input (0V-5V) from an external load sharing module (LSM) for paralleling of gensets
- Totalizers for engine starts, running hours, kWh, kVAh, etc.
- MODBUS over RS485 and CAN ports for communicating with other devices
- Supports parameter configuration at customer's site
- Offers multiple level passwords to avoid tampering in the configuration
- Access to each parameter through PC software or control keys of GC1100 controllers

Ordering Information:

GC1100 Controller SM0003913

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20˚C to 65˚C/ -30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V - 32V DC continuous; power transients as per ISO 7637 - 2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @ 12V or 24V DC

Crank-start Dropout Survival Period
50 mS for a drop from 12V DC

Charging Alternator Interface
0.25A @ 12V DC; 0.125A @ 24V DC
Diagnostic voltage measurement

Digital Output
6 x 0.75A high-side drivers configurable for pre-heat, hooter, etc

Digital Input
5 switch-to-ground inputs configurable for lube oil, temp, fuel level, etc

Analog Input
3 resistive inputs (10Ω - 1000Ω); fully configurable for various sources

Mains/Genset Voltage Measurement
20V – 350V AC RMS @ 5 – 75 Hz for Ph to N

Load Current Measurement
0A - 5A AC RMS for current transformer (CT) secondary

Magnetic Pickup Measurement
0.2V – 45V @ 10 Hz – 10 kHz

Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

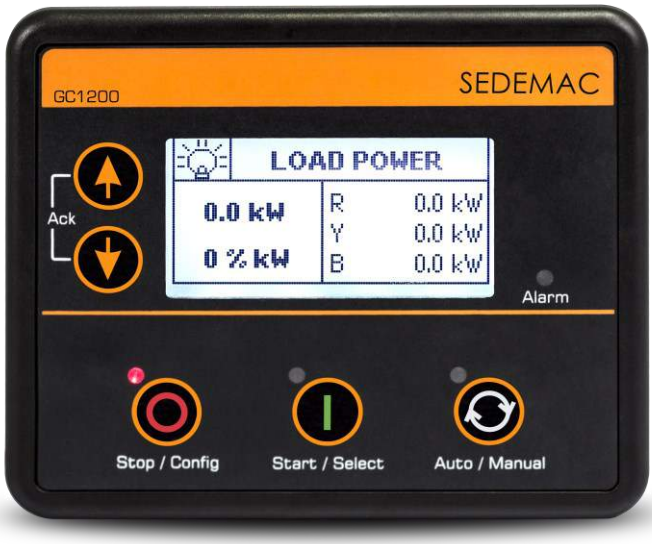
www.sedemac.com

GC1200

E-governing Genset Controller

HIGHLIGHTS »

- **Auto, manual and remote start/stop modes of operation**
- **Built-in electronic governor module**
- **MODBUS over RS485 and CAN port for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Offers multi-profile (10) in a single controller**



GC1200 Controller

SEDEMAC's GC1200 controllers are modern AMF genset controllers with user- friendly interface and full LCD graphics.

These controllers offer voltage and frequency measurement for mains as well as genset and electrical load measurement (true RMS). Prior to that they also offer engine safety monitoring feature

Deep Sleep Mode is a useful feature being provided to prolong the battery life by suspending the normal functions of the controller when DG is off.

MODBUS over **RS485** and **CAN ports are** provided for supporting the remote communication.

The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

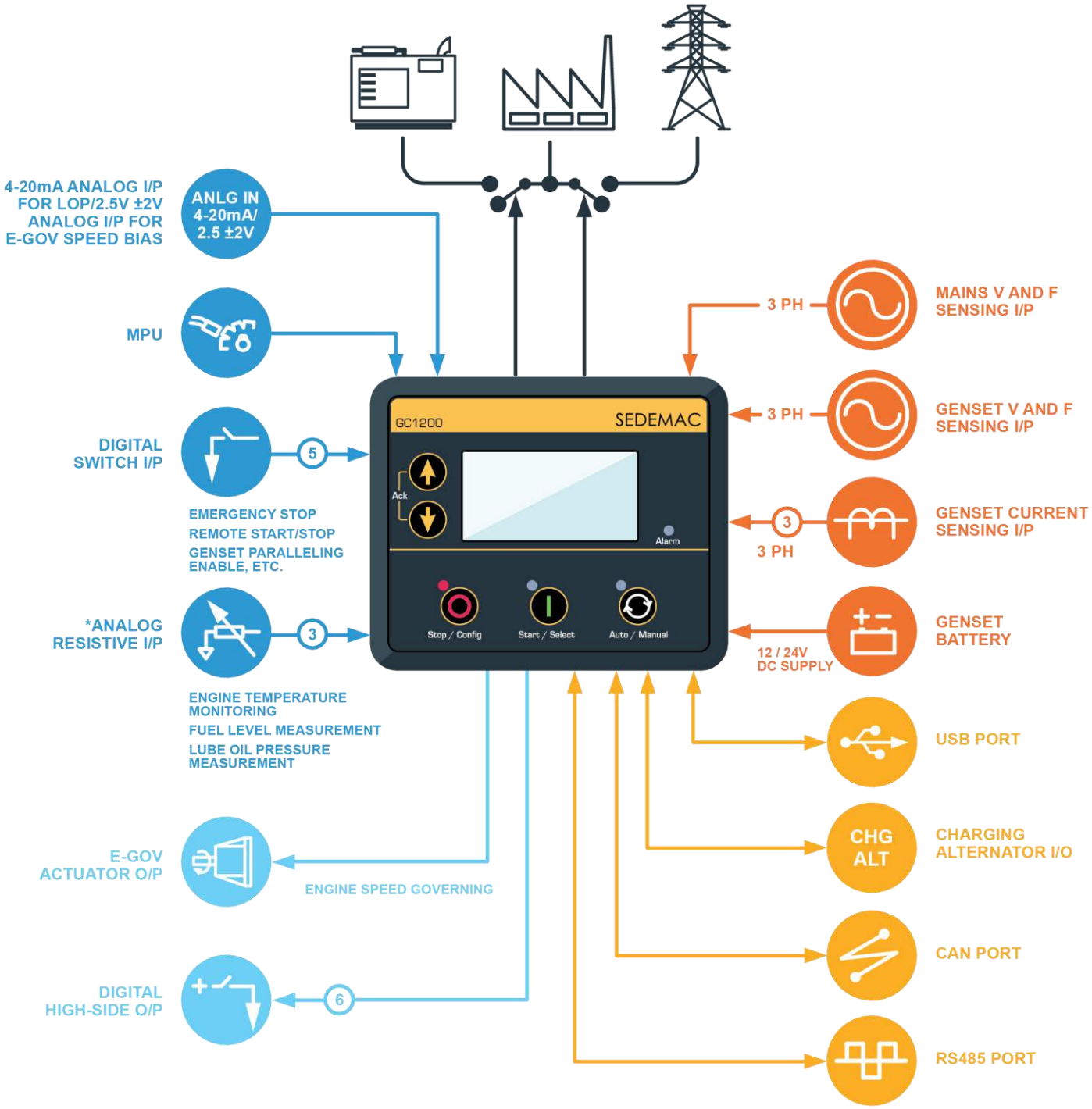
Real Time Clock (RTC) is provided for accurate event logging (100 counts) and data analysis.

Provision is there to Interface **Magnetic Pickup Unit** (MPU) to read the Engine RPM

In addition to above functionalities, GC1200 controller also offers **Electronic Governing** function with the help of a rotary actuator. It supports **genset paralleling** through an analog input signal of 2.5V ±2V .

GC1200

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics back-lit LCD display with power save mode
- Supports Auto (AMF, remote start/stop and cyclic modes) and manual modes
- Monitors 1-Ph/3-Ph voltage, frequency and load current for generator and mains; monitors 1-Ph/3-Ph power factor for generator
- Offers ''Auto exercise mode'' (2 events) to start and stop the genset for preconfigured time
- Monitors engine safety parameters through analog resistive sensors and digital input switches (e.g. - lube oil pressure, engine temperature, fuel level)
- Controls start relay, fuel relay, stop relay, alarm hooter and more
- Supports the battery charging alternator I/O interface
- MPU interface for engine speed measurement
- Versatile software capable of saving "events" during operation; event log for 100 events with real time clock (RTC) stamps and engine run hour information
- GC1200 has built-in electronic governor module which directly connects to SEDEMAC's rotary actuators* to guarantee speed governing performance as per ISO8528:G3 class
- Accepts target speed bias input (2.5V ±2V) from an external load sharing module (LSM) for paralleling of gensets
- Totalizers for engine starts, running hours, kWh, kVAh, etc.
- MODBUS over RS485 and CAN ports for communicating with other devices
- Supports parameter configuration at customer's site
- Offers multiple level passwords to avoid tampering in the configuration
- Access to each parameter through PC software or control keys of GC1200 controllers

Ordering Information:

GC1200 Controller SM0000048

GC1200 requires rotary actuator for E-governing

**Kindly refer SEDEMAC's datasheet of "rotary actuators for electronic speed governing" for interfacing an actuator with controller in governing application.*

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20˚C to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8V - 32V DC continuous; power transients as per ISO 7637 - 2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.08A @ 12V or 24V DC

Crank-start Dropout Survival Period
50 mS for a drop from 12V DC

Charging Alternator Interface
0.25A @ 12V DC; 0.125A @ 24V DC
Diagnostic voltage measurement

Digital Output
6 x 0.75A high-side drivers configurable for pre-heat, hooter, etc

Digital Input
5 switch-to-ground inputs configurable for lube oil, temp, fuel level, etc

Analog Input
3 resistive inputs (10Ω - 1000Ω); fully configurable for various sources
1 analog input (2.5V ± 2V) for E-Governor speed bias and paralleling

E-governor Module Interface
Output for rotary actuator
(2.5V ± 2V) input for target speed bias; 2.5V for zero bias

Mains/Genset Voltage Measurement
20V – 350V AC RMS @ 5 – 75 Hz for Ph to N

Load Current Measurement
0A - 5A AC RMS for current transformer (CT) secondary

Magnetic Pickup Measurement
0.2V – 45V @ 10 Hz – 10 kHz

Dimensions
External dimensions: 139 mm x 114 mm x 37.5 mm
Mounting panel cut: 118 mm x 93 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

GC2100

Feature-rich AMF Genset Controller

HIGHLIGHTS »

- **Monitoring of telecom site battery backup level and shelter temperature**
- **Cyclic start/stop operation mode tailor made for saving fuel in telecom tower applications**
- **AMF, manual and remote start/stop modes**
- **MODBUS over RS485 for remote communication**
- **Manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**
- **Real Time Clock (RTC) support for task scheduling and event log stamping**
- **Backlit, full graphics LCD with power saving feature**

GC2100 Controller

SEDEMAC's GC2100 is a modern AMF genset controller with user-friendly HMI and full graphics LCD. The controller has been designed keeping in mind the needs of the telecom tower application as well as non-telecom applications.

In addition to monitoring the engine safety & electrical load, GC2100 controller offers monitoring of **Site battery backup voltage and shelter temperature** for the telecom tower application to reduce the fuel consumption.

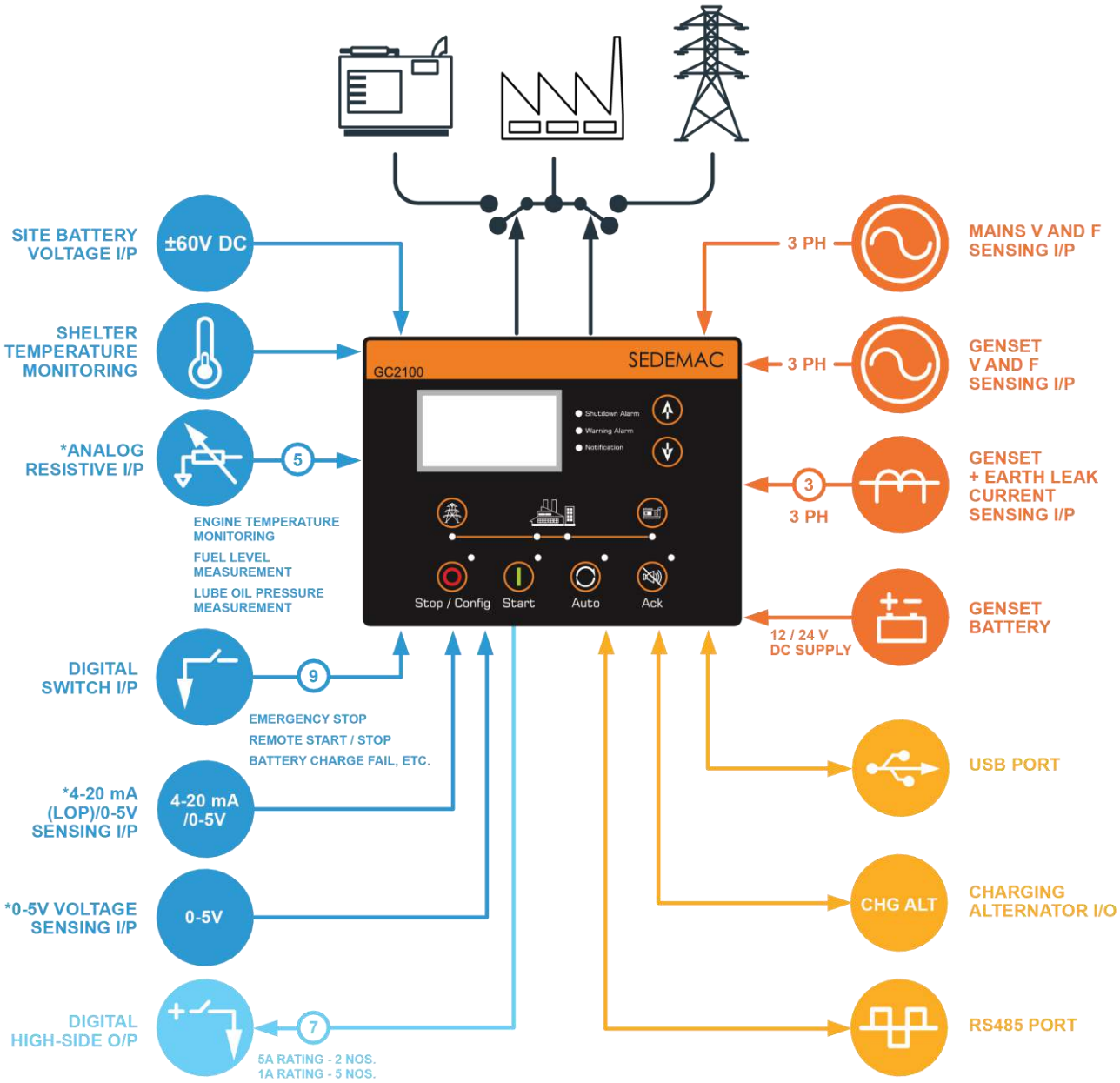
MODBUS over **RS485** is provided for supporting the remote communication.

The device has a **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

For task scheduling and event log stamping, the GC2100 also sports the **Real Time Clock** (RTC) feature. The controller also incorporates special power saving modes to prolong the genset battery backup when the engine is off.

GC2100

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics, backlit LCD with power save mode
- Supports Auto (site battery monitoring, AMF, remote start / stop modes, auto exercise and cyclic mode) and manual operation modes
- Monitors 1-ph/3-ph voltage, frequency, load current and power factor for generator
- Monitors engine safety parameters through both analog resistive sensors and digital switches (e.g. lube oil pressure, engine temperature, fuel level)
- Controls start relay, fuel relay, alarm hooter and more as "digital outputs"
- Monitors the telecom site battery backup level and shelter temperature to avoid unnecessary engine running and fuel expense at telecom tower sites
- Supports the battery charging alternator I/O interface
- Versatile software capable of saving "events" during operation; event log for 100 events with Real Time Clock (RTC) stamps and information regarding engine run hours
- Totalizers for engine starts, engine trips, engine run hours, genset kWh, kVAh, kVArh, Mains kWh, kVAh, kVArh
- Measures mains kW, kVA.
- MODBUS over RS485 is used for communication with other devices
- Configuration access to each parameter through PC software or control keys on the product fascia

Ordering information:

Controllers :

GC2100.....SM0003918

SPECIFICATIONS »

Environmental Compliance

Operating/Storage Temperature
-20˚C to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes for 8-500 Hz, as per IEC 60068-2-6

Humidity
0% to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8-32V DC continuous; power transients as per ISO 7637-2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.09A @ 12V or 24V

Crank-start Dropout Survival Period
50 mS for a drop from 12V
Able to survive 0 V for 50 mS

Charging Alternator Interface
0.25A @ 12V; 0.125A @ 24V
Diagnostic voltage measurement

Digital Output
5 x 1A high-side drivers configurable for pre-heat, hooter, ...
2 x 5A high-side drivers configurable for start relay, stop solenoid, fuel relay, contactors, etc.

Digital Input
9 x switch-to-ground inputs for lube oil, temp, fuel level, ...

Analog Input
5 x resistive inputs (10Ω - 5000Ω); fully configurable for various sources
1 x 4-20mA (LOP) / 0-5 V inputs
1 x 0-5 V inputs
1 x differential input (±60V DC) for site battery voltage

Mains/Genset Voltage Measurement
20V – 350V AC RMS @ 5 –75 Hz for Ph to N

Load Current Measurement
0A – 5A AC RMS for current transformer (CT) secondary

Dimensions
External dimensions: 207 mm x 160 mm x 38.5 mm
Mounting panel cut: 189 mm x 143 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

GC2599

AMF Controller for Variable / Fixed Speed AC Gensets

HIGHLIGHTS »

- **Built-in electronic governor module for variable speed governing at telecom tower sites**
- **Monitoring of telecom site battery level and shelter temperature**
- **Cyclic start/stop operation mode tailor made for saving fuel in telecom tower applications**
- **AMF, manual and remote start/stop mode with night restriction option**
- **Real time clock (RTC) support for task scheduling and event log stamping**
- **MODBUS over RS485 for remote communication & manually configurable via front fascia or via USB port using SEDEMAC's configuration utility**

GC2599 Controller

SEDEMAC's GC2599 is a modern AMF genset controller with user-friendly HMI and full graphics LCD. The controller has been designed keeping in mind the requirements of the telecom tower application as well as non-telecom applications.

A key feature in the GC2599 controller is load-based **variable speed operation** for telecom tower gensets. The controller incorporates a built-in governor module that directly drives SEDEMAC's **rotary actuators** and helps save fuel consumption under part load conditions prevalent in the telecom tower application. The governor module can also be configured to operate in **0% droop fixed speed operation** to support non-telecom applications with sensitivity to power quality.

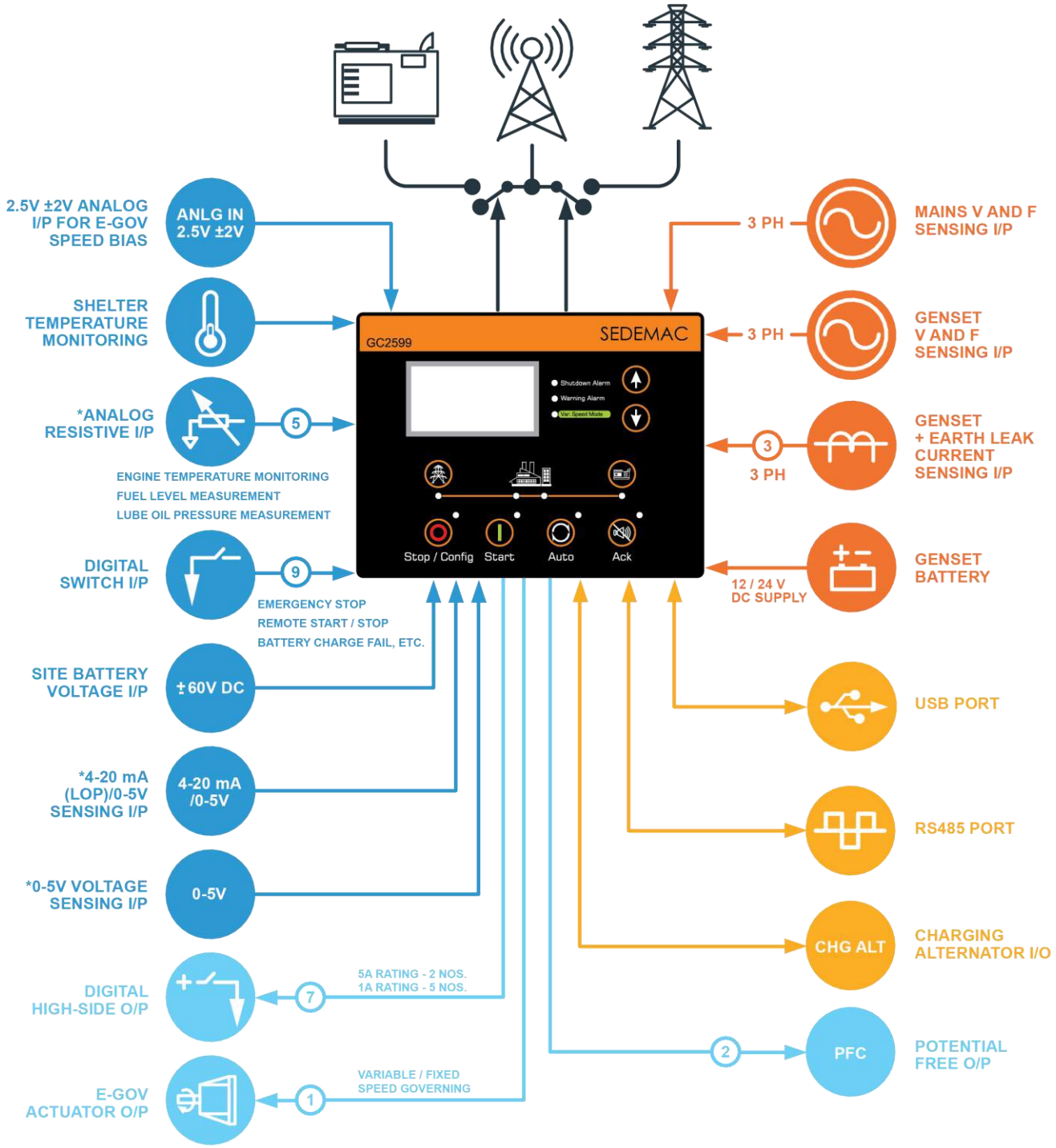
The GC2599 controller is packaged with a wide range of inputs and output capabilities. Monitoring of **Site battery voltage and shelter temperature** for the telecom tower application to reduce the fuel consumption and Potential-Free-Contact (PFC) outputs for contactor driving are two key highlights of the product.

MODBUS over **RS485** is provided for supporting the remote communication.

The device has an **USB** (Type-B) port that empowers it to communicate with SEDEMAC's proprietary Microsoft Windows based utility "**SEDEMAC-Smart Config**".

GC2599

SCHEMATIC »



*ANALOG INPUT CAN BE USED AS DIGITAL INPUT

FEATURES »

- Full graphics LCD with power save mode
- Supports Auto (site battery monitoring, AMF, remote start / stop modes, auto exercise and cyclic mode) and manual operation modes
- Built-in electronic governor module for fixed speed and variable speed operations; Directly connects to SEDEMAC's Rotary Actuators* to guarantee ISO8528:G3 class performance for 0% droop mode
- Monitors the telecom site battery backup level and shelter temperature to avoid unnecessary engine running and fuel expense at telecom tower sites
- Monitors 1-ph/3-ph voltage, frequency and load current for generator and mains; monitors 1-Ph/3-Ph power factor for generator
- Monitors engine safety parameters through both analog resistive sensors and digital switches (e.g. - lube oil pressure, engine temperature, fuel level)
- Controls start relay, fuel relay, alarm hooter and more as "digital outputs"
- Supports the battery charging alternator I/O interface
- Versatile software capable of saving "events" during operation; event log for 100 events with RTC stamps and engine run hour information
- Totalizers for engine starts, engine trips, engine run hour, genset kWh, kVAh, kVArh and mains kWh, kVAh, kVArh
- Measures mains kW, kVA.
- MODBUS over RS485 is used for communication with other devices
- Configuration access to each parameter through PC software or control keys on the product fascia

Ordering information:

GC2599.....SM0005239

Kindly refer SEDEMAC's datasheet of "Rotary Actuators for Electronic Speed Governing" for interfacing actuator with the controller in governing application

SPECIFICATIONS »

Environmental Testing Standards

Operating/Storage Temperature
-20˚C to 65˚C/-30˚C to 70˚C, as per IEC 60068-2-1, 2

Vibration
2G in X, Y and Z axes for 8-500 Hz, as per IEC 60068-2-6

Humidity
0 to 95% RH, as per IEC 60068-2-78

EMI/EMC
Compliance as per IEC 61000-6-2, 4

Power Supply
8-32V DC continuous; power transients as per ISO 7637-2

Ingress Protection
IP65 for front face with optional gasket, as per IEC 60529

Shock
15 g for 11 mS, compliance as per IEC 60068-2-27

Technical

Maximum Standby Current
~0.09A @ 12V or 24V; 1A (additional) for the actuator

Crank-start Dropout Survival Period
50 mS for a drop from 12V
Able to survive 0 V for 50 mS

Charging Alternator Interface
0.25A @ 12V; 0.125A @ 24V
Diagnostic voltage measurement

Digital Output
5 x 1A high-side drivers configurable for pre-heat, hooter, etc.
2 x 5A high-side drivers configurable for start relay, stop solenoid, fuel relay, etc
2 x 8A PFC outputs

Digital Input
9 x switch-to-ground inputs for lube oil pressure, temp, low fuel

Analog Input
5 x resistive inputs (10Ω - 5000Ω); fully configurable for various sources
1 x 4-20mA (LOP) / 0-5 V inputs
1 x 0-5 V inputs
1 x differential input (±60V DC) for site battery voltage

E-governor Module
1A, 2-phase output for SEDEMAC's rotary actuators
2.5 ± 2V input for target speed bias; 2.5V for zero bias

Mains / Genset Voltage Measurement
20V – 350V AC RMS @ 5 – 75 Hz for Ph to N

Load Current Measurement
0A – 5A AC RMS for current transformer (CT) secondary

Dimensions
External dimensions: 207 mm x 160 mm x 38.5 mm
Mounting panel cut: 189 mm x 143 mm

SEDEMAC GENSET CONTROLS »

Genset Controllers

Electronic Governors

ATS Controllers

R2D3 IIoT Platform

Battery Chargers

www.sedemac.com

About Company

SEDEMAC Mechatronics is a mechatronic products company engaged in the business of control solutions for automotive and off-highway powertrains. The company manufactures a wide range of control products like a variety of AMF genset controllers, electronic governors and variable speed controls for diesel gensets / engines. SEDEMAC's portfolio for gasoline engines / powertrains includes products like Smart Ignition, Integrated Starter Generator (ISG), ECU, Automated Manual Transmissions (AMT) ECU, Electronic Fuel Injection (EFI) ECU and so on.

A group of researchers founded SEDEMAC in 2007 in Mumbai, India. What started as a small lab-based enterprise has now grown into a rapidly growing company that supplies / licenses unique control products to the leading OEMs in India.

Each of the SEDEMAC's products goes through complete cycle of need of identification, development of relevant technology, design of practical products around developed technology, validation of products coordinating with key potential customers, organization of supply chain for series production, delivery to customers and effective after sales product support.

SEDEMAC is #1 in genset market control in India today and is the preferred choice of leading OEM's for unique control products.

Our Facilities

SEDEMAC MECHATRONICS PVT. LTD.
Pallod Farms, Baner Road, Pune 411 045
Phone: +91 20 67313500
Email: sales@sedemac.com
www.sedemac.com

SED-PDT-DTS-GC2599/1.00